A PROOF OF THEOREM 5.1

Theorem (Refined approximation of Collision-max to NMaxSim). *Let P be a set of m vectors drawn i.i.d. uniformly at random from the unit sphere $\mathbb{S}^{d-1}$, and let $q \in \mathbb{S}^{d-1}$ be a query. Assume repetition i of SimHash has signature length $k_{\text{sim}} = O(\frac{\log(m\delta^{-1})}{\varepsilon})$. Let*

$$P_{bucket} := \{p \in P : \phi_i(p) = \phi_i(q)\}, \qquad p^* := \arg\max_{p \in P} \langle q, p\rangle, \qquad \mathcal{A}_i := P_{bucket} \cup \{p^*\},$$

and denote $N := |\mathcal{A}_i|$. Assume $|P_{bucket}| \geq 1$ (equivalently $N \geq 2$). Then with probability at least $1-\delta$, the estimation errors satisfy

$$\text{NMaxSim}(q, P) - \bar{s}_i(q, P) \;\leq\; C \cdot \frac{\varepsilon\pi^2}{2\sqrt{2\log N}}, \qquad \text{NMaxSim}(q, P) - s_i(q, P) \;\leq\; \frac{\varepsilon\pi^2}{2},$$

where $C > 0$ is a constant depending only on the dimension d.

Proof. We fix the repetition index i and condition on the query q. Recall $g_{i,1}, \ldots, g_{i,k_{\text{sim}}} \sim \mathcal{N}(0, I_d)$ are the Gaussian vectors defining the k_{sim} SimHash hyperplanes in repetition i. We write the SimHash map as $\phi_i : \mathbb{S}^{d-1} \to \{0,1\}^{k_{\text{sim}}}$.

Step 1: Geometric localization via SimHash concentration. By Corollary H.1, setting $k_{\text{sim}} = \Theta(\log(m\delta^{-1})/\varepsilon)$, with high probability for every $p \in P_{\text{bucket}}$,

$$\theta(q, p) \leq \alpha \quad \Longrightarrow \quad \langle q, p\rangle \in [\cos\alpha, 1].$$

Since $P_{\text{bucket}} \neq \emptyset$, pick any $p_0 \in P_{\text{bucket}}$, so $\langle q, p_0\rangle \geq \cos\alpha$. By optimality of p^*,

$$\langle q, p^*\rangle \geq \langle q, p_0\rangle \geq \cos\alpha \quad \Longrightarrow \quad \theta(q, p^*) \leq \alpha,$$

hence for all $p \in \mathcal{A}_i$ we have $\langle q, p\rangle \in [\cos\alpha, 1]$.

Define the interval length

$$L := 1 - \cos\alpha.$$

Using $1 - \cos t \leq t^2/2$ for all t, we obtain

$$L = 1 - \cos\alpha \leq \alpha^2/2 = \pi^2\varepsilon/2. \tag{11}$$

Step 2: Reduce the error to the top gap in the augmented set. Let $\{X(p) = \langle q, p\rangle : p \in \mathcal{A}_i\}$ sorted as $X_{(1)} \geq X_{(2)} \geq \cdots$. Then

$$\text{NMaxSim}(q, P) = \langle q, p^*\rangle = X_{(1)}.$$

Define the collision-max estimator $\bar{s}_i(q, P) := \max_{p \in P_{\text{bucket}}} \langle q, p\rangle$. If $p^* \in P_{\text{bucket}}$, then $\bar{s}_i = X_{(1)}$ and the error is 0. Otherwise $\bar{s}_i = X_{(2)}$. In both cases,

$$\text{NMaxSim}(q, P) - \bar{s}_i(q, P) \leq X_{(1)} - X_{(2)}. \tag{12}$$

Step 3: Sub-Gaussianity of the similarities. Condition on the high-probability event from Step 1 (Corollary H.1), so that for every $p \in \mathcal{A}_i$ we have

$$X(p) = \langle q, p\rangle \in [\cos\alpha, 1] = [1 - L, 1], \qquad L := 1 - \cos\alpha.$$

Fix any $p \in \mathcal{A}_i$ and write $X := X(p)$. Let $\mu := \mathbb{E}[X]$ (under the current conditioning). Since X is almost surely bounded in an interval of length L, Hoeffding's lemma yields that for all $\lambda \in \mathbb{R}$,

$$\mathbb{E}\exp\big(\lambda(X - \mu)\big) \;\leq\; \exp\Big(\frac{\lambda^2 L^2}{8}\Big). \tag{13}$$

Equivalently, X is sub-Gaussian with proxy parameter

$$\sigma^2 = \frac{L^2}{4} \qquad \text{(i.e., } \sigma = L/2 \text{ in the convention } \mathbb{E}e^{\lambda(X-\mu)} \leq e^{\lambda^2\sigma^2/2}\text{)}.$$

Lemma A.1 (Gap between the top two sub-Gaussian order statistics). Let $X_1, \ldots, X_n$ be independent mean-zero sub-Gaussian random variables with common parameter $K > 0$, i.e.

$$\Pr(|X_i| \geq t) \leq 2\exp\Big(-\frac{t^2}{K^2}\Big), \qquad \forall\, t \geq 0,\ \forall\, i \in [n].$$

Let $X_{(1)} \geq X_{(2)} \geq \cdots \geq X_{(n)}$ be the order statistics and define $\Delta_n := X_{(1)} - X_{(2)}$. Then there exist universal constants $c, C > 0$ such that for all $u \geq 0$,

$$\Pr(\Delta_n \geq u) \;\leq\; \exp\left(-c\,\frac{u\sqrt{\log n}}{K}\right),$$

and consequently,

$$\mathbb{E}[\Delta_n] \;\leq\; C\,\frac{K}{\sqrt{\log n}}.$$

Proof. Fix $t \geq 0$. By the sub-Gaussian tail bound, for each i,

$$\Pr(X_i \geq t) \leq \exp\left(-\frac{t^2}{K^2}\right).$$

By a union bound,

$$\Pr(X_{(1)} \geq t) = \Pr\left(\bigcup_{i=1}^{n}\{X_i \geq t\}\right) \leq n \exp\left(-\frac{t^2}{K^2}\right).$$

Set $t_n := K\sqrt{2\log n}$. Then $\Pr(X_{(1)} \geq t_n) \leq ne^{-2\log n} = n^{-1}$, so $X_{(1)}$ concentrates around scale $K\sqrt{\log n}$.

Next, for any $u \geq 0$,

$$\{\Delta_n \geq u\} \subseteq \Big\{\exists\, i \neq j :\ X_i \geq t_n,\ X_j \leq t_n - u\Big\}.$$

Applying a union bound over ordered pairs (i, j) and independence,

$$\Pr(\Delta_n \geq u) \leq \sum_{i \neq j} \Pr(X_i \geq t_n)\Pr(X_j \leq t_n - u) \leq n(n-1)\Pr(X_1 \geq t_n)\Pr(X_1 \leq t_n - u).$$

Using $\Pr(X_1 \geq t_n) \leq \exp(-t_n^2/K^2) = n^{-2}$ and $\Pr(X_1 \leq t_n - u) = 1 - \Pr(X_1 > t_n - u) \leq \exp\big(-\Pr(X_1 > t_n - u)\big) \leq 1$, together with the sub-Gaussian tail $\Pr(X_1 > t_n - u) \leq \exp(-(t_n - u)^2/K^2)$, one obtains an exponential decay in u at rate proportional to $\sqrt{\log n}/K$, i.e.

$$\Pr(\Delta_n \geq u) \leq \exp\left(-c\,\frac{u\sqrt{\log n}}{K}\right)$$

for a universal constant $c > 0$ (absorbing numerical constants).

Finally, integrate the tail bound:

$$\mathbb{E}[\Delta_n] = \int_0^\infty \Pr(\Delta_n \geq u)\,du \leq \int_0^\infty \exp\left(-c\,\frac{u\sqrt{\log n}}{K}\right)du = C\,\frac{K}{\sqrt{\log n}},$$

for another universal constant $C > 0$. □

Step 4: High-probability bound on the top gap via the sub-Gaussian gap lemma. By Step 1, with probability at least $1 - \delta$, for every $p \in \mathcal{A}_i$ we have $X(p) = \langle q, p\rangle \in [1 - L, 1]$ with $L := 1 - \cos\alpha$. In what follows, we work on this high-probability event.

Step 4.1: Convert (13) **to the tail form in the lemma.** Fix any $p \in \mathcal{A}_i$ and write $X := X(p)$ and $\mu := \mathbb{E}[X]$. From (13) (Hoeffding's lemma), we have

$$\mathbb{E}\big[\exp(\lambda(X - \mu))\big] \leq \exp\Big(\frac{\lambda^2 L^2}{8}\Big), \qquad \forall \lambda \in \mathbb{R}.$$

By the standard Chernoff bound, this implies the two-sided tail

$$\Pr\big(|X - \mu| \geq t\big) \leq 2\exp\Big(-\frac{2t^2}{L^2}\Big) = 2\exp\Big(-\frac{t^2}{K^2}\Big), \qquad K := \frac{L}{\sqrt{2}}.$$

Thus, the centered similarities are sub-Gaussian in the sense of the lemma with parameter $K = L/\sqrt{2}$.

Step 4.2: Apply the expectation gap bound. Enumerate $\mathcal{A}_i = \{p_1, \ldots, p_N\}$ and set $X_j := \langle q, p_j\rangle$. Let $X_{(1)} \geq X_{(2)} \geq \cdots \geq X_{(N)}$ and $\Delta_N := X_{(1)} - X_{(2)}$. Note that subtracting a common mean does not change gaps, so Δ_N is the same for $\{X_j\}$ and $\{X_j - \mu\}$.

Applying Lemma (Gap between the top two sub-Gaussian order statistics) to $\{X_j - \mu\}_{j=1}^N$ yields universal $C > 0$ such that

$$\mathbb{E}[\Delta_N] \;\leq\; C\,\frac{K}{\sqrt{\log N}} \;=\; C\,\frac{L}{\sqrt{2\log N}}.$$

Step 4.3: Conclude the expected collision-max error bound. By (12), we have

$$\mathbb{E}\big[\mathrm{NMaxSim}(q, P) - \bar{s}_i(q, P)\big] \leq \mathbb{E}[\Delta_N] \leq C\,\frac{L}{\sqrt{2\log N}}.$$

Using (11) ($L \leq \pi^2\varepsilon/2$), we obtain

$$\mathbb{E}\big[\mathrm{NMaxSim}(q, P) - \bar{s}_i(q, P)\big] \leq C \cdot \frac{\varepsilon\pi^2}{2\sqrt{2\log N}}.$$

This bound holds with probability at least $1 - \delta$ (on the event from Step 1).

Step 5: Bound for the mean estimator $s_i(q,P)$. On $\mathcal{E}_i$, for every $p \in P_{\text{bucket}}$ we have $\theta(q,p) \le \alpha$, hence

$$1 - \cos(\theta(q,p)) \le 1 - \cos\alpha = L \le \frac{\varepsilon\pi^2}{2}.$$

Therefore,

$$\begin{aligned}\text{NMaxSim}(q,P) - s_i(q,P) &\le 1 - \frac{1}{|P_{\text{bucket}}|}\sum_{p\in P_{\text{bucket}}}\cos(\theta(q,p)) \\ &= \frac{1}{|P_{\text{bucket}}|}\sum_{p\in P_{\text{bucket}}}(1-\cos(\theta(q,p))) \\ &\le \frac{\varepsilon\pi^2}{2}.\end{aligned}$$

This completes the proof on $\mathcal{E}_i$, which holds with probability at least $1-\delta$. □

B PROOF OF THEOREM 5.2

Lemma B.1 (Max–mean gap under bounded inner products). Let $a_1,\dots,a_n \in [-1,1]$. Then

$$\max_{j\in[n]} a_j \;-\; \frac{1}{n}\sum_{j=1}^{n} a_j \;\le\; 2\Big(1-\frac{1}{n}\Big).$$

Proof. Let $a_{\max} = \max_j a_j \le 1$. Since $a_j \ge -1$, $\frac{1}{n}\sum_{j=1}^n a_j \ge \frac{1}{n}a_{\max} + \frac{n-1}{n}(-1)$. Rearranging gives $a_{\max} - \frac{1}{n}\sum_j a_j \le (1-\frac{1}{n})a_{\max} + \frac{n-1}{n} \le 2(1-\frac{1}{n})$. □

Theorem (Restatement of Theorem 5.2). *Assume all vectors are ℓ_2-normalized, so $\langle q,p\rangle \in [-1,1]$. Fix any repetition i, query Q, and document P. Define $n_{i,P}^{\max} := \max_{k\in[B]} n_{i,P}(k)$, $d_{i,P}^{\max} := \max\{1, n_{i,P}^{\max}\}$ Then the centroiding distortion satisfies*

$$0 \le \bar{s}_i(Q,P) - s_i(Q,P) \;\le\; \frac{2}{|Q|}\sum_{k=1}^{B} q_{i,Q}(k)\Big(1-\frac{1}{\max\{1,n_{i,P}(k)\}}\Big) \;\le\; 2\Big(1-\frac{1}{d_{i,P}^{\max}}\Big). \tag{14}$$

Proof. We first show $\bar{s}_i(Q,P) - s_i(Q,P) \le \frac{2}{|Q|}\sum_{k=1}^B q_{i,Q}(k)\Big(1-\frac{1}{\max\{1,n_{i,P}(k)\}}\Big)$. Fix a bucket k and a token q with $\phi_i(q)=k$. Let $\{p_j\}_{j=1}^n$ be the colliding document vectors, where $n = n_{i,P}(k)$. Then $m_i(q;P) = \max_{j\in[n]}\langle q,p_j\rangle$ and the centroid contribution equals $\frac{1}{n}\sum_{j=1}^n\langle q,p_j\rangle$ if $n>0$, and 0 if $n=0$. Applying Lemma B.1 to $a_j = \langle q,p_j\rangle$ yields

$$m_i(q;P) - \frac{1}{\max\{1,n\}}\sum_{j=1}^{n}\langle q,p_j\rangle \;\le\; 2\Big(1-\frac{1}{\max\{1,n\}}\Big).$$

Summing over all $q\in Q$ and grouping by buckets gives $0 \le \bar{s}_i(Q,P) - s_i(Q,P) \le U_i(Q,P) := \frac{2}{|Q|}\sum_{k=1}^B q_{i,Q}(k)\Big(1-\frac{1}{\max\{1,n_{i,P}(k)\}}\Big)$.

We next upper bound $U_i(Q,P)$ by a quantity independent of $q_{i,Q}(k)$. Let $d_k := \max\{1, n_{i,P}(k)\} \ge 1$ and $q_k := q_{i,Q}(k)$. Using $\sum_{k=1}^B q_k = |Q|$, rewrite

$$U_i(Q,P) = \frac{2}{|Q|}\sum_{k=1}^{B} q_k\Big(1-\frac{1}{d_k}\Big) = 2 - \frac{2}{|Q|}\sum_{k=1}^{B}\frac{q_k}{d_k}.$$

Define weights $w_k := q_k/|Q|$, so $w_k \ge 0$ and $\sum_{k=1}^B w_k = 1$. Then

$$U_i(Q,P) = 2 - 2\sum_{k=1}^{B}\frac{w_k}{d_k}.$$

Apply Engel (Titu) Cauchy–Schwarz:

$$\sum_{k=1}^{B}\frac{w_k}{d_k} = \sum_{k=1}^{B}\frac{w_k^2}{w_k d_k} \ge \frac{(\sum_{k=1}^B w_k)^2}{\sum_{k=1}^B w_k d_k} = \frac{1}{\sum_{k=1}^B w_k d_k}.$$

Since $\sum_{k=1}^B w_k d_k$ is a convex combination of $\{d_k\}$, we have $\sum_{k=1}^B w_k d_k \le \max_k d_k = d_{i,P}^{\max}$. Therefore,

$$\sum_{k=1}^{B}\frac{w_k}{d_k} \ge \frac{1}{d_{i,P}^{\max}} \quad\Longrightarrow\quad U_i(Q,P) \le 2 - 2\cdot\frac{1}{d_{i,P}^{\max}} = 2\Big(1-\frac{1}{d_{i,P}^{\max}}\Big).$$

Combining completes (14). □

C PROOF OF THEOREM 6.1

Theorem (Low-rank property of a single SimHash bucket sub-matrix). *Let $S := Q \cup P$ and $m := |S| = |Q| + |P|$. Fix a repetition index i and parameters $\varepsilon \in (0,1)$, $\delta \in (0,1)$. Assume all vectors hashed by SimHash are ℓ_2-normalized.*[1] *Draw $g_{i,1}, \ldots, g_{i,k_{\text{sim}}} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$ and form the SimHash map $\phi_i(\cdot) \in \{0,1\}^{k_{\text{sim}}}$ as in Lemma H.1. Let b be any bucket (i.e., any signature in $\{0,1\}^{k_{\text{sim}}}$), and let $X_{i,b} \in \mathbb{R}^{N_b \times d}$ be the sub-matrix that stacks the (normalized) vectors whose SimHash signatures equal b (see Equation (??)), where N_b is the number of vectors falling into bucket b in repetition i. Choose $k_{\text{sim}} = \Theta\left(\frac{\log(m\delta^{-1})}{\varepsilon}\right)$. Then, with probability at least $1-\delta$ over the randomness of $\{g_{i,j}\}_{j=1}^{k_{\text{sim}}}$, the following holds simultaneously for all buckets b:*

$$\operatorname{srank}(X_{i,b}) \;\le\; \frac{1}{\cos^2(\pi\sqrt{\varepsilon})}.$$

Proof. Let $\alpha := \pi\sqrt{\varepsilon}$. Apply Corollary H.1 to the set $S \times S$ (i.e., replace $Q \times P$ by $(Q \cup P) \times (Q \cup P)$; the proof is identical with $|S \times S| \le m^2$). With the stated choice of k_{sim}, we obtain an event $\mathcal{E}_i$ such that $\Pr(\mathcal{E}_i) \ge 1-\delta$ and, on $\mathcal{E}_i$, for all $x, y \in S$,

$$\phi_i(x) = \phi_i(y) \implies \theta(x,y) \le \alpha.$$

Now fix any bucket b and condition on $\mathcal{E}_i$. If $N_b \le 1$, then $\operatorname{srank}(X_{i,b}) \le 1$ holds trivially. Assume $N_b \ge 2$ and pick any row vector c of $X_{i,b}$ (so $\phi_i(c) = b$). For every row u_j of $X_{i,b}$ we also have $\phi_i(u_j) = b$, hence $\theta(u_j, c) \le \alpha$ and, since $\|u_j\|_2 = \|c\|_2 = 1$,

$$u_j^\top c \;\ge\; \cos\alpha.$$

We bound the stable rank $\operatorname{srank}(X_{i,b}) = \|X_{i,b}\|_F^2 / \|X_{i,b}\|_2^2$:

(i) Frobenius norm. Because each row has unit norm,

$$\|X_{i,b}\|_F^2 \;=\; \sum_{j=1}^{N_b} \|u_j\|_2^2 \;=\; N_b.$$

(ii) Spectral norm. Using the unit test vector c,

$$\|X_{i,b}\|_2^2 \;=\; \sigma_1(X_{i,b})^2 \;\ge\; \|X_{i,b}c\|_2^2 \;=\; \sum_{j=1}^{N_b} (u_j^\top c)^2 \;\ge\; \sum_{j=1}^{N_b} \cos^2\alpha \;=\; N_b \cos^2\alpha.$$

Combining (i) and (ii) yields

$$\operatorname{srank}(X_{i,b}) \;=\; \frac{\|X_{i,b}\|_F^2}{\|X_{i,b}\|_2^2} \;\le\; \frac{N_b}{N_b \cos^2\alpha} \;=\; \frac{1}{\cos^2(\pi\sqrt{\varepsilon})}.$$

Since the argument holds for an arbitrary bucket b on the same event $\mathcal{E}_i$, the bound holds simultaneously for all buckets with probability at least $1-\delta$. □

Lemma C.1 (From Bucket-wise Low-rankness to the Full FDE Matrix). Let $X \in \mathbb{R}^{N \times (RBd_{\text{proj}})}$ be the full document FDE matrix, written as the horizontal concatenation of its bucket sub-matrices:

$$X \;=\; \Big[\, X_{1,1} \,\|\, X_{1,2} \,\|\, \cdots \,\|\, X_{i,b} \,\|\, \cdots \,\|\, X_{R,B} \,\Big],$$

where each $X_{i,b} \in \mathbb{R}^{N \times d_{\text{proj}}}$ stacks the vectors from bucket b in repetition i. Then the stable rank satisfies

$$\operatorname{srank}(X) \;\le\; \sum_{i=1}^{R} \sum_{b=1}^{B} \operatorname{srank}(X_{i,b}).$$

In particular, if for all (i,b) we have $\operatorname{srank}(X_{i,b}) \le \tau$, then

$$\operatorname{srank}(X) \;\le\; RB\,\tau.$$

Proof. By definition, $\operatorname{srank}(X) = \|X\|_F^2 / \|X\|_2^2$. For horizontal concatenation, the Frobenius norm decomposes:

$$\|X\|_F^2 \;=\; \sum_{i=1}^{R} \sum_{b=1}^{B} \|X_{i,b}\|_F^2.$$

Moreover, the spectral norm is monotone with respect to taking submatrices, hence

$$\|X\|_2 \;\ge\; \|X_{i,b}\|_2 \qquad \text{for every } (i,b).$$

[1] Equivalently, SimHash depends only on direction; normalization is without loss of generality.

Therefore,

$$\operatorname{srank}(X) = \frac{\sum_{i,b} \|X_{i,b}\|_F^2}{\|X\|_2^2} \le \sum_{i,b} \frac{\|X_{i,b}\|_F^2}{\|X_{i,b}\|_2^2} = \sum_{i,b} \operatorname{srank}(X_{i,b}),$$

which proves the claim. The final bound follows immediately if each $\operatorname{srank}(X_{i,b}) \le \tau$. □

D PROOF OF THEOREM 6.2

For each repetition $i \in [R]$, let (ϕ_i, ψ_i) be the random design used to build the i-th FDE block, and define

$$F^{(i)}(X; \phi_i, \psi_i) := \widetilde{\psi}_i\big(u_i(X, \phi_i)\big) \in \mathbb{R}^D, \quad F(X; \phi, \psi) := \big(F^{(1)}, \dots, F^{(R)}\big) \in \mathbb{R}^{d_{\mathrm{FDE}}}.$$

We emphasize that MUVERA samples (ϕ_i, ψ_i) once per repetition and then keeps them fixed for all data points; our result below is an *average over this random initialization.*

Theorem D.1 (Seed-averaged block-diagonality). Under the MUVERA random design facts (F1)–(F2), define the seed-averaged covariance

$$\bar{\Sigma} := \operatorname{Cov}_{X,\phi,\psi}\big[F(X; \phi, \psi)\big].$$

Then for any $i \ne j$,

$$\operatorname{Cov}_{X,\phi,\psi}\big[F^{(i)}(X; \phi_i, \psi_i),\ F^{(j)}(X; \phi_j, \psi_j)\big] = \mathbf{0},$$

and hence $\bar{\Sigma}$ is block-diagonal:

$$\bar{\Sigma} = \operatorname{diag}(\bar{\Sigma}_1, \dots, \bar{\Sigma}_R), \qquad \bar{\Sigma}_i := \operatorname{Cov}_{X,\phi_i,\psi_i}\big[F^{(i)}(X; \phi_i, \psi_i)\big].$$

Proof. Fix $i \ne j$. Write

$$F_i := F^{(i)}(X; \phi_i, \psi_i) = \widetilde{\psi}_i(u_i), \qquad F_j := F^{(j)}(X; \phi_j, \psi_j) = \widetilde{\psi}_j(u_j),$$

where $u_i := u_i(X, \phi_i)$ and $u_j := u_j(X, \phi_j)$. Let $Z := (X, \phi_i, \phi_j)$. Conditioned on Z, the vectors u_i, u_j are deterministic.

We apply the law of total covariance:

$$\operatorname{Cov}(F_i, F_j) = \mathbb{E}\big[\operatorname{Cov}(F_i, F_j \mid Z)\big] + \operatorname{Cov}\big(\mathbb{E}[F_i \mid Z],\ \mathbb{E}[F_j \mid Z]\big).$$

We show both terms are zero.

Step 1: conditional means vanish. By (F1), ψ_i (hence $\widetilde{\psi}_i$) is independent of Z. Therefore, conditioning on Z fixes u_i and we can take expectation only over ψ_i:

$$\mathbb{E}[F_i \mid Z] = \mathbb{E}_{\psi_i}\Big[\widetilde{\psi}_i(u_i)\Big].$$

By the zero-mean property (F2) with $v = u_i$ (deterministic under Z), $\mathbb{E}[F_i \mid Z] = \mathbf{0}$. Similarly, $\mathbb{E}[F_j \mid Z] = \mathbf{0}$. Hence

$$\operatorname{Cov}\big(\mathbb{E}[F_i \mid Z],\ \mathbb{E}[F_j \mid Z]\big) = \operatorname{Cov}(\mathbf{0}, \mathbf{0}) = \mathbf{0}.$$

Step 2: conditional cross-covariance vanishes. By definition,

$$\operatorname{Cov}(F_i, F_j \mid Z) = \mathbb{E}[F_i F_j^\top \mid Z] - \mathbb{E}[F_i \mid Z]\, \mathbb{E}[F_j^\top \mid Z].$$

The second term is $\mathbf{0}$ by Step 1. It remains to show $\mathbb{E}[F_i F_j^\top \mid Z] = \mathbf{0}$.

Conditioned on Z, u_i, u_j are deterministic. By (F1), ψ_i and ψ_j are independent (and independent of Z), hence $\widetilde{\psi}_i$ and $\widetilde{\psi}_j$ are independent given Z. Therefore,

$$\mathbb{E}[F_i F_j^\top \mid Z] = \mathbb{E}\Big[\widetilde{\psi}_i(u_i)\, \widetilde{\psi}_j(u_j)^\top \mid Z\Big] = \mathbb{E}\Big[\widetilde{\psi}_i(u_i) \mid Z\Big]\ \mathbb{E}\Big[\widetilde{\psi}_j(u_j)^\top \mid Z\Big].$$

Using Step 1 again, each conditional mean is $\mathbf{0}$, so $\mathbb{E}[F_i F_j^\top \mid Z] = \mathbf{0}$. Hence $\operatorname{Cov}(F_i, F_j \mid Z) = \mathbf{0}$ and thus

$$\mathbb{E}\big[\operatorname{Cov}(F_i, F_j \mid Z)\big] = \mathbf{0}.$$

Conclusion. Both terms in the total-covariance decomposition are $\mathbf{0}$, so $\operatorname{Cov}(F_i, F_j) = \mathbf{0}$ for all $i \ne j$. This implies $\bar{\Sigma}$ is block-diagonal with diagonal blocks $\bar{\Sigma}_1, \dots, \bar{\Sigma}_R$. □

E PROOF OF THEOREM 7.1

F PROOF OF THEOREM 8.1

We now address the validity of applying a PCA transformation matrix learned solely from the corpus (documents) to unseen query vectors. The core intuition rests on the **Low Stable Rank** property (Theorem C) combined with the geometry of SimHash.

(A3) Dense Corpus Assumption. We assume the corpus P is sufficiently large such that for the given number of buckets B, every bucket is non-empty. That is, for every repetition i and bucket b, the set $P_b^{(i)} = \{p \in P : \phi_i(p) = b\}$ is non-empty, and thus the aggregate centroid $c_b := u_b^{(i)}(P)$ is well-defined.

We define the *Corpus Subspace* $\mathcal{S}_{\text{doc}}$ as the subspace spanned by the centroids of all buckets:

$$\mathcal{S}_{\text{doc}} \;=\; \text{span}\left(\{c_1, \ldots, c_B\}\right) \subset \mathbb{R}^d.$$

In a PCA context, since the intra-bucket stable rank is close to 1 (Theorem C), the variance of the data is dominated by the directions of these centroids. Therefore, the principal components learned from the corpus will align with $\mathcal{S}_{\text{doc}}$.

Theorem F.1 (Query Coverage by Corpus-PCA). Let $q \in \mathbb{R}^d$ be an arbitrary unit-norm query vector. Let Π_{doc} be the orthogonal projection operator onto the Corpus Subspace $\mathcal{S}_{\text{doc}}$. Under the conditions of Theorem C, if q falls into bucket b (i.e., $\phi_i(q) = b$), then the reconstruction error is bounded by:

$$\|q - \Pi_{\text{doc}} q\|_2 \;\leq\; \sin\alpha,$$

where α is the collision angle threshold defined in Theorem C.

Proof. The proof relies on the geometric constraints imposed by the SimHash collision.

Step 1: Angular proximity to the centroid. Since q falls into bucket b, it shares the same hash signature as all document vectors $p \in P_b^{(i)}$.By the derivation in Theorem C, with high probability, the angle between any two vectors in the same bucket is bounded by α.Since the centroid c_b is a convex combination of vectors in $P_b^{(i)}$, it lies within the angular cone spanned by the bucket.Conservatively, we guarantee that the angle between the query q and the bucket centroid c_b satisfies:

$$\theta := \angle(q, c_b) \;\leq\; \alpha.$$

Step 2: Projection onto the subspace. By definition, the centroid $c_b \in \mathcal{S}_{\text{doc}}$.Since Π_{doc} projects onto $\mathcal{S}_{\text{doc}}$, it preserves c_b perfectly: $\Pi_{\text{doc}} c_b = c_b$.We decompose the query vector q relative to the direction of c_b. Let $\hat{c}_b = c_b / \|c_b\|_2$. We can write:

$$q \;=\; (\cos\theta)\hat{c}_b \;+\; (\sin\theta) v_\perp,$$

where $v_\perp$ is a unit vector orthogonal to $\hat{c}_b$.The projection error is:

$$\|q - \Pi_{\text{doc}} q\|_2 \;=\; \|(I - \Pi_{\text{doc}})q\|_2.$$

Substituting the decomposition:

$$(I - \Pi_{\text{doc}})q \;=\; (I - \Pi_{\text{doc}})\left((\cos\theta)\hat{c}_b + (\sin\theta)v_\perp\right).$$

Since $\hat{c}_b \in \mathcal{S}_{\text{doc}}$, $(I - \Pi_{\text{doc}})\hat{c}_b = 0$. Thus:

$$\|(I - \Pi_{\text{doc}})q\|_2 \;=\; \|(\sin\theta)(I - \Pi_{\text{doc}})v_\perp\|_2 \;\leq\; |\sin\theta| \cdot \|v_\perp\|_2 \cdot \|I - \Pi_{\text{doc}}\|_2.$$

Since projections are non-expansive, $\|I - \Pi_{\text{doc}}\|_2 \leq 1$. Given $\|v_\perp\| = 1$ and $\theta \leq \alpha$, we obtain:

$$\|q - \Pi_{\text{doc}} q\|_2 \;\leq\; \sin\alpha.$$

□

G THEORETICAL ANALYSIS

Notation. We use a generic multi-vector object X (either a query Q or a document P), represented as a finite set of ℓ_2-normalized vectors

$$X \;=\; \{x_1, \ldots, x_m\} \subset \mathbb{R}^d.$$

The fixed-dimensional encoding (FDE) of X has the form

$$F(X) \;=\; \left(F^{(1)}(X) \parallel F^{(2)}(X) \parallel \cdots \parallel F^{(R)}(X)\right) \in \mathbb{R}^{d_{\text{FDE}}},$$

with $R = R_{\text{reps}}$ repetitions and per-repetition block $F^{(i)}(X) \in \mathbb{R}^D$, where $D = B \cdot d_{\text{proj}}$ and $d_{\text{FDE}} = R \cdot D$.

In each repetition $i \in [R]$, we first draw a random partition

$$\phi_i : \mathbb{R}^d \to [B] = \{1, \ldots, B\}$$

and assign each vector $x \in X$ to one of B buckets. For bucket $b \in [B]$ we define the aggregated vector

$$u_b^{(i)}(X) \;:=\; Aggregate_{x \in X:\, \phi_i(x) = b} \;\; x \in \mathbb{R}^d,$$

the Aggregate operator is sum for queries and average for corpus. We then apply an inner random projection $\psi_i : \mathbb{R}^d \to \mathbb{R}^{d_{\text{proj}}}$ to each $u_b^{(i)}(X)$ and concatenate:

$$F^{(i)}(X) \;:=\; \Big(\psi_i\big(u_1^{(i)}(X)\big) \,\|\, \cdots \,\|\, \psi_i\big(u_B^{(i)}(X)\big)\Big) \in \mathbb{R}^{B\cdot d_{\text{proj}}}.$$

Different repetitions $\{(\phi_i, \psi_i)\}_{i=1}^R$ are sampled independently.

Inner projection. Throughout we instantiate ψ_i as a Rademacher Johnson–Lindenstrauss (JL) projection:

$$\psi_i(x) \;=\; \frac{1}{\sqrt{d_{\text{proj}}}} S_i x, \quad (S_i)_{ab} \overset{\text{i.i.d.}}{\sim} \text{Rad}(\pm 1).$$

Block-Wise Inner projection. For repetition $i \in [R]$, recall that the bucketwise aggregates are

$$u_b^{(i)}(X) \;:=\; \text{Aggregate}_{x \in X:\, \phi_i(x)=b}\; x \in \mathbb{R}^d, \qquad b \in [B].$$

We stack them into a single vector

$$u_i(X, \phi_i) \;:=\; \big(u_1^{(i)}(X) \,\|\, \cdots \,\|\, u_B^{(i)}(X)\big) \in \mathbb{R}^{Bd}.$$

We extend the inner projection $\psi_i : \mathbb{R}^d \to \mathbb{R}^{d_{\text{proj}}}$ to a blockwise JL map

$$\widetilde{\psi}_i : \mathbb{R}^{Bd} \to \mathbb{R}^{Bd_{\text{proj}}}, \qquad \widetilde{\psi}_i\big(v_1 \,\|\, \cdots \,\|\, v_B\big) := \big(\psi_i(v_1) \,\|\, \cdots \,\|\, \psi_i(v_B)\big),$$

where each ψ_i is instantiated as a Rademacher JL projection. With this notation,

$$F^{(i)}(X) \;=\; \widetilde{\psi}_i\big(u_i(X, \phi_i)\big) \in \mathbb{R}^D, \qquad D = B\, d_{\text{proj}}.$$

G.1 Facts and Assumptions

We separate (i) *design facts* dictated by the MUVERA construction from (ii) *modeling assumptions* on the data/aggregation needed by our analysis.

(F1) MUVERA random design (facts from construction). For each repetition $i \in [R]$, the pair (ϕ_i, ψ_i) is sampled once and then kept fixed for all data points. Moreover, $\{(\phi_i, \psi_i)\}_{i=1}^R$ are independent across i, and the random design is independent of the data object X.

(F2) Centered Rademacher Projection (fact). Each inner projection is instantiated as $\psi_i(x) = \frac{1}{\sqrt{d_{\text{proj}}}} S_i x$ with i.i.d. Rademacher entries (± 1). Hence for every deterministic vector v,

$$\mathbb{E}_{\psi_i}[\psi_i(v)] = \mathbf{0}, \qquad \text{and therefore} \qquad \mathbb{E}_{\psi_i}[\widetilde{\psi}_i(v)] = \mathbf{0}.$$

(A1) ℓ_2-normalized input vectors. For every multi-vector object $X = \{x_1, \ldots, x_m\} \subset \mathbb{R}^d$, all base vectors are ℓ_2-normalized:

$$\|x_j\|_2 = 1, \qquad j = 1, \ldots, m.$$

(A2) Corpus-side bucket aggregation and empty buckets. For the corpus (documents), the bucket aggregate is the *average*:

$$u_b^{(i)}(X) \;:=\; \frac{1}{|X_b^{(i)}|} \sum_{x \in X_b^{(i)}} x, \qquad X_b^{(i)} := \{x \in X : \phi_i(x) = b\},$$

and we set $u_b^{(i)}(X) = \mathbf{0}$ when $X_b^{(i)} = \emptyset$. Under (A1), this implies the uniform bound

$$\|u_b^{(i)}(X)\|_2 \le 1 \qquad \text{for all } b \in [B],\ i \in [R],\ X.$$

(Indeed, $u_b^{(i)}(X)$ lies in the convex hull of the unit ball.)

H BOUND-OPTIMAL REPETITION SELECTION FOR MUVERA-STYLE FDES

H.1 Background: From multi-vector similarity to fixed-dimensional encodings

Multi-vector similarity (normalized Chamfer). A late-interaction retriever represents a query/document as finite sets of ℓ_2-normalized vectors $Q, P \subset \mathbb{R}^d$, and compares them by the normalized Chamfer similarity

$$\text{NCHAMFER}(Q, P) \;:=\; \frac{1}{|Q|} \sum_{q \in Q} \max_{p \in P} \langle q, p \rangle \in [-1, 1]. \tag{15}$$

MUVERA-style FDE via repeated randomized partitions. The key idea of MUVERA is to approximate (15) by a dot product between fixed-dimensional encodings. Each *repetition* $i \in [R]$ samples an independent randomized partition $\phi_i : \mathbb{R}^d \to [B]$ (e.g., via SimHash), buckets query/document vectors by ϕ_i, and aggregates within each bucket.

Document-side fill_empty_clusters (no empty buckets). To avoid systematic loss when a query bucket contains no document vectors, MUVERA applies `fill_empty_clusters` on the *document side* only: for each repetition i and bucket $k \in [B]$, define a fill representative

$$p_i^{\text{fill}}(k) \in \arg\min_{p \in P} \|\phi_i(p) - k\|_0, \tag{16}$$

where $\|\cdot\|_0$ denotes Hamming distance between bucket codes (for SimHash buckets). Ties are broken arbitrarily. Importantly, we do *not* fill query buckets, to avoid duplicating query contributions across buckets.

Bucket-count notation. For repetition i, define query bucket counts and document bucket counts

$$q_{i,Q}(k) \;:=\; \big|\{q \in Q : \phi_i(q) = k\}\big| \quad n_{i,P}(k) := \big|\{p \in P : \phi_i(p) = k\}\big|$$

One repetition: bucketed sum (query) and centroid-or-fill (document). Fix repetition i. For each bucket $k \in [B]$, define the aggregated vectors

$$\tilde{q}_i(k) \;=\; \sum_{q \in Q:\phi_i(q)=k} q, \;\; \tilde{p}_{i,b}(P) = \begin{cases} \frac{1}{n_{i,b}(P)} \sum\limits_{p \in P:\phi_i(p)=b} p, & n_{i,b}(P) > 0, \\ \tilde{p}_{i,b'}(P), & n_{i,b}(P) = 0, b' = \operatorname{argmax}_{b'} \|\mathrm{Bit}(p) - \mathrm{Bit}(b)\|_0 \end{cases} \tag{17}$$

where $n_{i,P}(k) := \big|\{p \in P : \phi_i(p) = k\}\big|$ is the document bucket occupancy. The flattened concatenation $\tilde{q}_i := \big(\tilde{q}_i(1)\| \cdots \|\tilde{q}_i(B)\big)$ and $\tilde{p}_i := \big(\tilde{p}_i(1)\| \cdots \|\tilde{p}_i(B)\big)$ yields a fixed-length vector in $\mathbb{R}^{Bd}$ for this repetition.

Per-repetition score, and averaging across repetitions. Define the normalized block score

$$s_i(Q, P) \;:=\; \frac{1}{|Q|} \langle \tilde{q}_i, \tilde{p}_i \rangle. \tag{18}$$

H.1.1 SimHash Concentration.

Lemma H.1 (SimHash: Hamming distance concentrates around the angle). Fix a repetition index i. Let $g_{i,1}, \ldots, g_{i,k_{\text{sim}}} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$ and define the k_{sim}-bit SimHash $\phi_i(x) \in \{0,1\}^{k_{\text{sim}}}$ by

$$\phi_i(x) := \big(\phi_{i,1}(x), \ldots, \phi_{i,k_{\text{sim}}}(x)\big), \qquad \phi_{i,j}(x) := \mathbf{1}\big(\langle g_{i,j}, x \rangle > 0\big).$$

For any $q \in Q, p \in P$, let $\theta(q,p) \in [0,\pi]$ denote the angle between them and $\mathrm{Ham}(\phi_i(q), \phi_i(p))$ the Hamming distance between their signatures, setting $k_{\text{sim}} = O\left(\frac{\log(m^2/(\varepsilon\delta))}{\varepsilon}\right) = O\left(\frac{\log(m\delta^{-1})}{\varepsilon}\right)$, then with high probablity

$$\Pr\left(\left|\mathrm{Ham}(\phi_i(q), \phi_i(p)) - k_{sim} \cdot \frac{\theta(q,p)}{\pi}\right| > \sqrt{\varepsilon} k_{sim}\right) \le (\frac{\varepsilon\delta}{m^2}).$$

Proof. For each bit $j \in [k_{\text{sim}}]$, define the disagreement indicator

$$Z_j := \mathbf{1}\big(\phi_{i,j}(q) \neq \phi_{i,j}(p)\big) = \mathbf{1}\big(\mathrm{sign}\langle g_{i,j}, q \rangle \neq \mathrm{sign}\langle g_{i,j}, p \rangle\big),$$

so that $\mathrm{Ham}(\phi_i(q), \phi_i(p)) = \sum_{j=1}^{k_{\text{sim}}} Z_j$. Since $\{g_{i,j}\}$ are i.i.d., $\{Z_j\}$ are independent and $Z_j \in \{0,1\}$.

Step 1: Compute $\mathbb{E}[Z_j] = \theta(q,p)/\pi$. Let $U := \mathrm{span}\{q,p\}$ (at most two-dimensional). Write $g_{i,j} = g^{\parallel} + g^{\perp}$ with $g^{\parallel} \in U$ and $g^{\perp} \perp U$. Because $q, p \in U$, we have $\langle g_{i,j}, q \rangle = \langle g^{\parallel}, q \rangle$ and $\langle g_{i,j}, p \rangle = \langle g^{\parallel}, p \rangle$, hence the event $\{Z_j = 1\}$ depends only on the 2D vector $g^{\parallel}$.

By rotational symmetry of the standard Gaussian, the direction of $g^{\parallel}$ (after normalization) is uniform on the unit circle in U. Therefore, we may equivalently sample a random direction u uniformly on that circle and consider the signs of $\langle u, q \rangle$ and $\langle u, p \rangle$. In U, the condition $\langle u, q \rangle > 0$ corresponds to an open semicircle (arc length π) centered at q, and similarly for p. The signs differ iff u lies in the symmetric difference of these two semicircles. When the angle between q and p equals $\theta = \theta(q,p)$, this symmetric difference consists of two arcs of length θ each, hence total arc length 2θ. Since the full circle has arc length 2π,

$$\Pr(Z_j = 1) = \frac{2\theta}{2\pi} = \frac{\theta}{\pi},$$

so $\mathbb{E}[Z_j] = \theta(q,p)/\pi$.

Step 2: Concentration via Hoeffding's inequality. The variables $Z_j \in [0,1]$ are independent with mean θ/π. Hoeffding's inequality states that for any $t > 0$,

$$\Pr\left(\left|\sum_{j=1}^{k_{\text{sim}}} Z_j - k_{\text{sim}} \cdot \frac{\theta}{\pi}\right| > t\right) \le 2\exp\left(-\frac{2t^2}{k_{\text{sim}}}\right).$$

Setting $t = \sqrt{\varepsilon}\, k_{\text{sim}}$ gives

$$\Pr\left(\left|\text{Ham}(\phi_i(q), \phi_i(p)) - \frac{k_{\text{sim}}\,\theta}{\pi}\right| > \sqrt{\varepsilon}\, k_{\text{sim}}\right) \leq 2\exp(-2\varepsilon k_{\text{sim}}) \tag{19}$$

$$\leq (\frac{\varepsilon\delta}{m^2}) \tag{20}$$

where we took $k \geq 1/2 \cdot \log(\frac{m^2}{\varepsilon\delta})/\varepsilon$, which completes the proof. □

Corollary H.1 (Collision implies small angle (high probability)). *Fix a repetition index i and $\varepsilon \in (0,1)$. Setting $k_{sim} = O\left(\frac{\log(m\delta^{-1})}{\varepsilon}\right)$ Then with probability at least $1 - \varepsilon\delta$ over the randomness of $\{g_{i,j}\}_{j=1}^{k_{\text{sim}}}$, the following holds simultaneously for all $(q,p) \in Q \times P$:*

$$\phi_i(q) = \phi_i(p) \implies \theta(q,p) \leq \pi\sqrt{\varepsilon} =: \alpha.$$

Proof. Define the event

$$\mathcal{E}_i := \left\{\forall (q,p) \in Q \times P : \left|\text{Ham}(\phi_i(q), \phi_i(p)) - k_{\text{sim}} \cdot \frac{\theta(q,p)}{\pi}\right| \leq \sqrt{\varepsilon}\, k_{\text{sim}}\right\}.$$

By Lemma H.1, for any fixed pair (q,p),

$$\Pr\left(\left|\text{Ham}(\phi_i(q), \phi_i(p)) - k_{\text{sim}} \cdot \frac{\theta(q,p)}{\pi}\right| > \sqrt{\varepsilon} k_{\text{sim}}\right) \leq \frac{\varepsilon\delta}{m^2}.$$

Applying the union bound over all $|Q \times P| \leq m^2$ pairs yields

$$\Pr(\mathcal{E}_i^c) \;\leq\; \sum_{(q,p)\in Q\times P} \frac{\varepsilon\delta}{m^2} \;\leq\; \varepsilon\delta,$$

hence $\Pr(\mathcal{E}_i) \geq 1 - \varepsilon\delta$.

Now condition on $\mathcal{E}_i$. If $\phi_i(q) = \phi_i(p)$ then $\text{Ham}(\phi_i(q), \phi_i(p)) = 0$, and under $\mathcal{E}_i$ we get

$$k_{\text{sim}} \cdot \frac{\theta(q,p)}{\pi} \;\leq\; \sqrt{\varepsilon}\, k_{\text{sim}},$$

so $\theta(q,p) \leq \pi\sqrt{\varepsilon} = \alpha$. □